LUMAFLOW · STEM CHECK

LumaFlow STEM check

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/01b-stem-check.md @ 8a57af4		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	S34 closing route as decided by Amish (O-4 (a), 2026-10-25); no verdict changed
0.1	2026-10-25	Amish Chadha	First STEM check (kit 1.8 upgrade of a TRL 3 repo): 39 claims on option A, 6 each on options B and C; four detail failures of option A fixed under Amish's standing instruction (LMF-DEC-001 D-S1 to D-S4), two of them found by the independent review, which is recorded

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Verdict: the working principle of option A, the existing design (a flow-switched UV-C LED reactor that shines along a reflective, water-filled tube and stops the water when its wall sensor reads too little light), sits within known science. Of the 51 rows, **39 are within known science and 12 are plausible, to confirm**. Four details of option A failed and were fixed in the same session under Amish's standing instruction of 2026-10-23. The first run found two: UV-C refracted outward in the quartz window fell on the rubber (EPDM) window gasket and sensor window washer (S9; both are now expanded PTFE), and the retaining ring screws, at the torque the build plan gave, bent the 2 mm stainless ring to about its proof stress (S12; now anti-seize and 0.21 N·m). The independent review found two more: with the normally closed valve downstream of the flow sensor, opening the faucet could never start the unit (S37; an outlet pressure sensor now starts it), and the alarm threshold, set from an unweighted dose, let the true dose fall to about 37 mJ/cm² and missed a failed LED (S38; now set from the 275 nm weighted dose). Seven of option A's plausible rows need measurement or the chosen parts' datasheets, not calculation: photorepair (S6), regrowth while the water stands (S7), light through the PTFE onto the acetal cap (S10), the valve's flow coefficient (S22), LED derating (S33, worst case 37.5 mJ/cm²), mixing in the reactor (S34) and the photodiode's spectral response (S39). So `stem_check` stays **open**; the repo keeps its TRL 3 under STANDARDS section 20, and the dose margin question is with Amish (LMF-DEC-001, O-3); the mixing question was decided by Amish on 2026-10-25 (O-4 (a): keep the side inlet, tracer test at TRL 4). Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which imports the calculation note's script `docs/04-calcs/sizing.py` (LMF-CAL-001), so the rows rerun whenever the model changes.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv` by `stem_check.py`). Rows S1 to S39 are option A (S33 to S39 were added after the independent review); B1 to B6 option B; C1 to C6 option C.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Of the 0.36 W of UV-C from the LEDs, 84.8 % enters the water and 52.9 % is absorbed by it at 90 %/cm	Physics	Optics and radiation	Conservation of radiant power in the Monte Carlo ray trace (absorbed + lost = entering)	in 0.3052 W, out 0.3052 W	Within known science	LMF-CAL-001 B2, B3
S2	Reduction equivalent dose (RED) at 1.2 L/min and 90 %/cm reaches 40 mJ/cm2 for a 254 nm-equivalent response (laminar profile)	Physics	Optics and radiation	Fluence rate = absorbed power density / alpha; dose = fluence rate x residence time along each streamline; RED for a 20 mJ/cm2-per-log challenge organism	50.9 mJ/cm2 >= 40 mJ/cm2	Within known science	LMF-CAL-001 B5
S3	Organism response at 275 nm is close enough to that at 254 nm that the RED stays above 40 mJ/cm2	Biology	Microbiology	Fluence-based rate constants of MS2: 280 nm LED 0.052 against LP lamp 0.056 cm2/mJ (ratio 0.93, taken for 275 nm though measured at 280 nm); E. coli 0.31 against 0.27 (Beck et al. 2017); RED x 0.93	47.3 mJ/cm2 >= 40 mJ/cm2	Within known science	LMF-CAL-001 B10 (margin 18 %; PR-002 at risk)
S4	Even the least-dosed streamline (laminar, on the axis) gets the 4-log dose for Cryptosporidium and Giardia	Biology	Microbiology	40 CFR 141.720(d)(1): 22 mJ/cm2 for 4-log Cryptosporidium and Giardia at 254 nm	38.4 mJ/cm2 >= 22 mJ/cm2	Within known science	LMF-CAL-001 B7 x 0.93
S5	LumaFlow is not a virus barrier: 40 to 51 mJ/cm2 is far below the 186 mJ/cm2 for 4-log virus (adenovirus)	Biology	Microbiology	40 CFR 141.720(d)(1) virus column: 39 mJ/cm2 for 0.5 log, 186 for 4 log	Non-claim kept in PRC, PRS and the safety notes; about 0.5 log of adenovirus	Within known science	LMF-PRC-001 Safety; PRS PR-019

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					at the design dose		
S6	Bacteria do not regain viability by photoreactivation or dark repair after the design dose	Biology	Microbiology	Photorepair of UV-C damage is known in bacteria and is smaller at higher doses and at LED wavelengths near 280 nm (literature review in Beck et al. 2017); not quantified for this reactor	No calculation possible; plate counts with light and dark holding at TRL 4	Plausible, to confirm at TRL 3	Open
S7	Water left standing in the reactor between draws does not regrow bacteria past the outlet, with the idle LED pulse of 10 s every 4 h	Biology	Microbiology	Biofilm regrows in stagnant point-of-use devices; the idle pulse gives the standing water about 38 mJ/cm2 volume-average per 10 s pulse (twice the 5 s run-on dose of LMF-CAL-001 D4; corrected after the independent review); downstream valve and tubing are not dosed	Plate counts at TRL 4 (decided 2026-10-02)	Plausible, to confirm at TRL 3	Open
S8	The dose target is met even if the liner reflects only 0.80, as plain PTFE might, not 0.95	Physics	Optics and radiation	Ray trace rerun at a wetted reflectance of 0.80, times the 275 nm factor 0.93; reflectance of wetted sintered PTFE at 275 nm to be measured (the reviewer's trace needs	42.1 mJ/cm2 >= 40 mJ/cm2	Within known science	LMF-CAL-001 B8 x 0.93 (5 % margin)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				about 0.72 or more)			
S9	No UV-C falls on an elastomer: window gasket and sensor window washer	Chemistry	Materials and polymer chemistry	Photo-oxidation of EPDM under UV-C; ray trace of the light refracted outward in the window: 4.2 % of the LED output, 2.6 mW/cm2, about 856 J/cm2 a year on the gasket annulus	First run: EPDM in the light path (failed). Fix: both seals in food-contact expanded PTFE, which UV-C does not degrade	Within known science	LMF-DEC-001 D-S1 (2026-10-25)
S10	UV-C passing through the 3 mm PTFE wall and top disc is too weak to degrade the acetal upper cap	Chemistry	Materials and polymer chemistry	Sintered PTFE is a diffuse reflector but not opaque at 3 mm; acetal yellows and embrittles under UV; diffuse transmittance of the liner grade at 275 nm is not published for this part	Measure the liner's transmittance at TRL 4; fallback: a 0.3 mm 316 foil cup in the liner pocket	Plausible, to confirm at TRL 3	Open
S11	Quartz window 57 x 10 mm on a 51 mm seat stays below the 6.8 MPa design stress at 8 bar	Physics	Strength of materials	Simply supported circular plate, sigma = 3 (3 + nu) p a^2 / (8 t^2); fused quartz design stress 6.8 MPa (Momentive)	6.18 MPa <= 6.8 MPa	Within known science	LMF-CAL-001 G4
S12	The 2 mm 316 retaining ring stays within 2/3 of its proof stress at the specified screw torque	Physics	Strength of materials	Ring as a cantilever strip between the screw circle (65 mm) and the washer centre (54 mm), arm 5.5 mm; preload = T / (K d), K 0.13 to 0.16	131 MPa <= 137 MPa	Within known science	LMF-DEC-001 D-S2 (2026-10-25): 316 sheet, anti-seize, 0.21 N m; worst nut factor 0.13

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				with anti-seize; first run at 0.5 N m dry: 202 MPa (failed); 0.3 N m dry was 121 to 162 MPa (reviewer: too narrow a window)			
S13	At 8 bar the window gasket keeps at least twice the water pressure as seating stress	Physics	Mechanics and statics	Least preload 2.62 kN (0.21 N m, K 0.16) less the pressure load on the 50 mm seal (1.57 kN) over the 588 mm2 gasket	1.79 MPa >= 1.6 MPa	Within known science	LMF-DEC-001 D-S2
S14	M5 316 tie rods carry the end load at 8 bar within 0.6 of their proof stress	Physics	Strength of materials	Stress = end load / (4 x 14.2 mm2); A4-70 proof stress 450 MPa (ISO 3506-1)	47 MPa <= 270 MPa	Within known science	LMF-CAL-001 G7
S15	Reactor tube hoop stress at 8 bar is a small share of the 316 proof stress	Physics	Strength of materials	Thin-wall hoop stress p r / t; allowable 205 / 3	8.3 MPa <= 68 MPa	Within known science	LMF-CAL-001 G6
S16	Acetal roof over the liner pocket stays below its long-term design stress at 8 bar	Physics	Strength of materials	Clamped circular plate, sigma = 0.75 p a^2 / t^2; acetal long-term design stress 10 MPa (assumed)	2.8 MPa <= 10 MPa	Within known science	LMF-CAL-001 G8
S17	LED board stays at or below 50 degC with continuous flow, at the worst film coefficient	Physics	Heat transfer	Two thermal paths in parallel: finned sink to 30 degC air, spreader and 316 cap to 25 degC water	45.1 degC <= 50 degC	Within known science	LMF-CAL-001 F3
S18	LED junctions stay below an 85 degC rating	Engineering and technology	Power and link budgets	Board temperature + 10 K/W x 2.11 W per LED; 85	62 degC <= 85 degC	Within known science	LMF-CAL-001 F4

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				degC is a common junction limit for 3535 UV-C LEDs, to confirm on the chosen part			
S19	The water warms by about 0.11 K carrying 8.9 W of LED heat	Physics	Thermodynamics	Q = m_dot cp dT at 1.2 L/min	claimed 0.11 K, estimate 0.107 K (ratio 1.03)	Within known science	LMF-CAL-001 F2
S20	Thermal shock from 5 degC water on a warm window does not crack fused quartz	Physics	Heat transfer	Fused silica expansion 0.55e-6 /K (Momentive); thermal stress E alpha dT / (1 - nu) for a 40 K step with E 72 GPa	1.9 MPa <= 6.8 MPa	Within known science	This check
S21	Flow in the 50 mm bore is laminar at 1.2 L/min (the profile itself is not known: see S34)	Physics	Fluid mechanics	Re = rho U D / mu with D 50 mm; transition near 2,300	507 <= 2300	Within known science	LMF-CAL-001 A4
S22	Pressure drop stays below 0.5 bar at 1.2 L/min	Physics	Fluid mechanics	dp = (Q / Kv)^2 per component: 0.15 bar on the assumed Kv 0.35 and 0.25, 0.39 with 0.2 and 0.15, 0.59 with a direct-acting valve of Kv 0.10 (independent review)	Depends on datasheet Kv; PR-008 at risk until the sensor and valve are chosen	Plausible, to confirm at TRL 3	Open
S23	Mains power while water flows stays at or below 25 W	Physics	Electromagnetism and circuits	LED 6 x 350 mA x 6.2 V through a 90 % driver, valve 4.8 W, controller 0.6 W, adapter 88 %	22.6 W <= 25 W	Within known science	LMF-CAL-001 E1
S24		Physics	Electromagnetism and circuits	Flow sensor 75 mW, outlet	0.23 W <= 0.5 W		LMF-CAL-001 E2

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	Standby power stays at or below 0.5 W			pressure sensor 10 mW and MCU sleep through a 70 % light-load adapter, plus 0.1 W no-load		Within known science	
S25	The wall photodiode signal at the alarm threshold is well above the amplifier noise floor	Engineering and technology	Sensing resolution	Photocurrent at S/S_ref = 0.79 against 10 pA (a hundred times a SiC photodiode's dark current of about 0.1 pA through a transimpedance amplifier)	11.35 nA >= 0.01 nA	Within known science	LMF-CAL-001 C1, C3
S26	LEDs switch on within 0.5 s of flow starting at 0.3 L/min	Engineering and technology	Control	First pulse of a 15 Hz per L/min sensor plus 20 ms	0.24 s <= 0.5 s	Within known science	LMF-CAL-001 D2
S27	LED running hours do not limit life within 10 years of household use	Engineering and technology	Power and link budgets	92 h a year (flow plus idle pulse) x 10 years against a 10,000 h rated life (L70 class, to confirm on the chosen part)	920 h <= 1e+04 h	Within known science	LMF-CAL-001 E4
S33	The dose stays above 40 mJ/cm2 with the LEDs derated for junction temperature and bin	Engineering and technology	Power and link budgets	Nominal 47.3 mJ/cm2 (B10); with about 12 % less output near 62 degC and a 10 % low bin: 37.5 (B11); both factors typical, not yet from the chosen LED's datasheet	Closes when the LED is chosen; if the stack holds, routes to Amish (LMF-DEC-001 O-3)	Plausible, to confirm at TRL 3	Open
S34	Mixing in the reactor leaves the RED between the plug and laminar limits	Physics	Fluid mechanics	The laminar profile never develops (entrance length 1.3 m against 0.23 m), the inlet jet has a port Reynolds number of about	Residence-time distribution by tracer test at TRL 4 with the side inlet kept (O-4 (a), decided by Amish	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				3,000 to 6,000 and the heated lower cap drives buoyant mixing (Richardson number about 6 to 12); a back-mixed reactor of 5 to 10 tanks in series would give 31 to 39 mJ/cm2 (reviewer's trace)	2026-10-25); perforated PTFE inlet ring ready if the test shows strong back-mixing		
S35	The water-film coefficient in the stainless lower cap bore is at least the 150 W/(m2 K) low end used for the board temperature	Physics	Heat transfer	Developing laminar flow over the first 18 mm (Sieder-Tate, Nu = 1.86 Gz^(1/3), Gz = 9973); the fully developed value Nu 3.66 (h about 44) the reviewer used applies only far downstream	486 W/(m2 K) >= 150 W/(m2 K)	Within known science	This check; LMF-CAL-001 F3
S36	Closing the outlet valve at full flow keeps the reactor within the 8 bar window rating	Physics	Fluid mechanics	Joukowsky surge rho c v in the 1 m PE inlet tube (c about 400 m/s, v 0.63 m/s) added to the 4 bar limiter setting; a rigid copper line would add more, so the tube from the tee stays PE (BOM line 15)	6.5 bar <= 8 bar	Within known science	This check; PR-017
S37	Opening the faucet starts the unit	Engineering and technology	Control	First run: the normally closed valve sits downstream of the flow sensor, so with the faucet open and the valve shut no	Start-up sequence in LMF-CAL-001 D5; standby 0.23 W	Within known science	LMF-DEC-001 D-S3 (2026-10-25)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				water flows and the flow sensor can never wake the controller (failed). Fix: an outlet pressure sensor between valve and faucet; the drop in line pressure starts the LEDs, then the valve opens and the flow sensor must confirm flow			
S38	One failed LED of six trips the low-dose alarm	Engineering and technology	Sensing resolution	The wall sensor sees the six LEDs about equally: S/S_ref = 5/6 against the alarm threshold, now set from the 275 nm weighted RED (40 / 47.3)	0.833 S/S_ref <= 0.846 S/S_ref	Within known science	LMF-DEC-001 D-S4 (2026-10-25)
S39	The SiC photodiode reads the UV-C, not the LEDs' longer-wavelength tail	Physics	Optics and radiation	SiC (band gap about 3.26 eV) has little response above about 380 nm; AlGaN LEDs emit a weak parasitic band near 300 to 400 nm that water hardly absorbs	Check against the chosen LED's spectrum and the photodiode's response curve	Plausible, to confirm at TRL 3	Open
S28	Average dose of 75.4 mJ/cm2 at 90 %/cm and 1.2 L/min	Mathematics	Units and dimensional analysis	Volume-average dose = absorbed power / (alpha Q), independent of geometry (the ray trace counts only the water between the ports, so it reads lower)	claimed 75.4 mJ/cm2, estimate 90.5 mJ/cm2 (ratio 0.83)	Within known science	LMF-CAL-001 B5
S29		Mathematics					This check

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	Monte Carlo noise in the RED is under 3 %		Numerical method	Second run of 600,000 photons with another seed	0.55 % <= 3 %	Within known science	
S30	No mercury anywhere in the unit	Chemistry	Environmental and atmospheric chemistry	AlGaN UV-C LEDs contain no mercury (Minamata Convention context in PRB)	Bill of materials review: no lamp, no mercury	Within known science	BOM lines 1 to 19
S31	Every wetted material is suitable for food contact: 316, PTFE and expanded PTFE, fused silica, EPDM O-rings out of the light, acetal; and the bought parts on the water path (flow sensor, valve, pressure sensor, tee, tube, restrictor) in food-contact, lead-free grades	Chemistry	Materials and polymer chemistry	21 CFR 177.1550 (perfluorocarbon resins), 177.2470 (acetal copolymer), 177.2600 (rubber articles); NSF/ANSI 61 certification to assess for the chosen grades	Reactor reviewed in the model; bought water-path parts to confirm at purchase (DEC To confirm); acetal is poorer in chloraminated water, which the well-water first user does not have	Within known science	PRS PR-011; REG G2, G3
S32	No galvanic cell between the aluminium heat path and the 316 cap	Chemistry	Electrochemistry and batteries	Aluminium and 316 touch only through the thermal pads on the dry side; the reactor's wetted metal is all 316, and the valve, flow sensor and pressure sensor are specified with plastic or stainless wetted parts (no brass)	Design review of the joint stack	Within known science	BOM line 5
B1	A 0.8 m2 hollow-fibre cartridge passes 1.2 L/min at no more than 0.5 bar	Physics	Fluid mechanics	Darcy flow through the membrane: flux 90 L/(m2 h) at 0.5 bar needs a	To confirm on a cartridge datasheet; a larger	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	transmembrane pressure			clean-water permeability of 180 L/(m2 h bar); PVDF and PES UF membranes span roughly 100 to 500	cartridge if not		
B2	Intact 0.01 to 0.02 um fibres remove bacteria (0.5 um and up) and protozoan cysts (3 um and up) by size	Biology	Microbiology	Size exclusion; US EPA Membrane Filtration Guidance Manual (EPA 815-R-06-009) credits log removal of Cryptosporidium only with direct integrity testing	Pore size 25 times smaller than the smallest bacteria	Within known science	Open
B3	Viruses (20 to 100 nm) are only partly removed	Biology	Microbiology	Virus sizes overlap the pore size; removal depends on the membrane grade	Not credited in scoring	Plausible, to confirm at TRL 3	Open
B4	A 1 bar air pressure-decay test finds a broken fibre without forcing air through intact pores	Physics	Fluid mechanics	Young-Laplace bubble point 4 gamma / d for a 0.02 um pore (ASTM D6908 practice)	0.1 MPa <= 14.4 MPa	Within known science	Open
B5	The slim 10 in housing holds 8 bar	Engineering and technology	Standards practice	Catalogue housings are commonly rated near 8.6 bar (125 psi); to confirm on the chosen housing	Datasheet check	Plausible, to confirm at TRL 3	Open
B6	Treatment needs no power; fouling shows as a rising gap between the two gauges	Physics	Fluid mechanics	Supply pressure drives the flow; TMP rises as the cake layer grows (Darcy)	No calculation needed	Within known science	Open
C1		Physics					Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	Twelve LEDs give an average dose of about 100 mJ/cm2 at 90 %/cm and 1.2 L/min		Optics and radiation	Absorbed share 29 % from single-pass absorption across the tube and a reflector of 0.80; average dose = absorbed power / (alpha Q); RED to come from a ray trace	claimed 100 mJ/cm2, estimate 97.8 mJ/cm2 (ratio 1.02)	Within known science	
C2	The RED of the radial reactor reaches 40 mJ/cm2	Physics	Optics and radiation	Laminar flow in a 20 mm tube (Re about 1,270) gives a reactor efficiency near 0.5, so RED about half the average	Ray trace needed before C could be chosen	Plausible, to confirm at TRL 3	Open
C3	The quartz flow tube holds 8 bar	Physics	Strength of materials	Hoop stress p r / t against the 6.8 MPa design stress	4.4 MPa <= 6.8 MPa	Within known science	Open
C4	LED strips stay at or below 50 degC with flow, cooled by their fins and through the end fittings into the water	Physics	Heat transfer	25.3 W through air (2.2 K/W) and water (0.76 K/W) paths in parallel from 25 degC water	39 degC <= 50 degC	Within known science	Open
C5	The seals at the tube ends stay out of UV-C guided along the quartz wall	Chemistry	Materials and polymer chemistry	Light trapped in the quartz wall by total internal reflection reaches its ends; PTFE-encapsulated O-rings there	By design in the option C model	Within known science	Open
C6	The same flow switch, dose photodiode and valve logic as option A work across a thin radial column	Engineering and technology	Control	The photodiode in the reflector gap sees direct LED light as well as light through the water, so its signal depends less on UVT than option A's;	To assess if C is chosen	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				mapping to assess			

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. A plausible row that only a measurement can close is carried to TRL 4 in the project plan (LMF-PLN-001).

Fixes made under the standing instruction

Row	What failed	Fix (LMF-DEC-001)	Result
S9	The retaining ring lets light into the window only within 51 mm, but rays refracted outward in 10 mm of quartz reach the window's top face out to about 34 mm radius. The EPDM gasket (25 to 28.5 mm) sits there, and its inner edge is the bottom of the water channel: about 4.2 % of the LED output, 2.6 mW/cm², roughly 860 J/cm² a year at 92 LED hours. EPDM is not UV-C stable, against PR-011. The sensor window washer sat in the same position behind its window	D-S1: window gasket and sensor window washer in food-contact expanded PTFE sheet of the same size. Options weighed: a PTFE-enveloped gasket (rubber core still sees light through thin PTFE), a metal mask ring (metal on quartz, a second interface), expanded PTFE (chosen: same geometry, compressible, UV-C stable). About USD 2 (estimate)	Row closed; model, BOM lines 9 and 17, build plan, PRC and PRS updated
S12, S13	The build plan said 0.5 N·m dry on the six M3 ring screws: about 5.0 kN, which bends the 2 mm 316 ring at 202 MPa (proof 205, allowable 137) and loads the PTFE washer near 10 MPa, where it creeps. The first fix, 0.3 N·m dry, left too narrow a window: the reviewer showed a nut factor of 0.15 overstresses the ring and 0.25 unseats the gasket	D-S2: A4 screws with anti-seize (nut factor about 0.13 to 0.16) at 0.21 N·m, ring cut from 316 sheet (not 316L). Options weighed: a 3 mm ring (moves the window up, changes the optics), Belleville washers (more parts), lubricated threads at lower torque (chosen). No cost	Ring at most 131 MPa, gasket at least 1.79 MPa at 8 bar; build plan 3.5, step 4 and a re-torque check after the first fill
S37	Found by the reviewer: in standby the normally closed outlet valve is shut and sits downstream of the flow sensor, so opening the faucet cannot make water flow and the flow sensor can never wake the controller. The design as drawn could not start	D-S3: an outlet pressure sensor on a push-fit tee between valve and faucet; the pressure drop when the faucet opens switches the LEDs on, the valve opens 20 ms later, and the flow sensor must confirm flow in 0.5 s or the valve shuts (fault). Options weighed: valve open in standby (4.8 W), latching valve (weaker fail-safe), faucet switch (user-facing), pressure sensor (chosen). About USD 15 (estimate), standby 0.23 W	Row closed; new BOM line 20, model, PRC, CAL D5 and E2, build plan
S38	Found by the reviewer: the alarm threshold S/S_ref = 0.79 came from the unweighted RED (50.9), so the true weighted dose at the alarm point was about 37 mJ/cm², and one failed LED (S/S_ref 0.83) would not trip it	D-S4: threshold from the 275 nm weighted RED (47.3): S/S_ref = 0.85. A firmware parameter, no cost	One failed LED trips the alarm; the UV level bar shows two of five segments in design water

None of the fixes changes the working principle, the pitch or the scope. All are recorded in LMF-DEC-001 as decided under Amish's standing instruction of 2026-10-23, and he can reverse them. The reviewer's points that would change a requirement Amish set or the design beyond a detail (the dose margin once the LED is chosen, an inlet flow distributor) went to Amish (LMF-DEC-001, O-3 and O-4); O-4 was decided on 2026-10-25 (side inlet kept, distributor held ready).

Fields not relevant

None. Every field is touched: physics (optics, strength of materials, heat transfer, fluid mechanics, circuits), chemistry (polymer stability under UV-C, food contact, galvanic couples, mercury), biology (microbial inactivation, photorepair, regrowth), mathematics (units, Monte Carlo noise) and engineering (sensing, control, power and life budgets, standards practice).

Concept options

- **Option A, axial UV-C LED reactor:** S1 to S39. Within known science after the four fixes; plausible rows S6, S7, S10, S22, S33, S34 and S39 wait for datasheets or TRL 4 measurements.
- **Option B, hollow-fibre ultrafilter:** within known science in principle (size exclusion, B2; integrity test well below the bubble point, B4). Plausible: membrane permeability for 1.2 L/min at 0.5 bar (B1), virus removal (B3, not credited) and the housing pressure rating (B5). Not dropped.
- **Option C, radial UV-C LED reactor:** within known science for dose order of magnitude (C1), quartz tube stress (C3), cooling through the water (C4) and seals out of the light (C5); its RED (C2) and sensor mapping (C6) need a ray trace. It draws about 39 W from the mains while water flows, which misses PR-009 (a requirement, not a science failure; scored in LMF-OPT-001). Not dropped.

Independent review

On 2026-10-25 a second agent (a separate Claude Code session, run headless with read-only access to a folder holding only this STEM check, the calculation note LMF-CAL-001, the product requirements LMF-PRS-001 and `sizing.py`, and none of the author's reasoning) was asked to refute the headline results and every "Within known science" verdict. It rebuilt the ray trace from the stated optics and reproduced the note (84.8 % entering, 52.3 % absorbed, RED 50.0 laminar and 73.8 plug against 50.9 and 74.9). It could not fetch the cited literature, so it checked arithmetic, not the sources. What it found and what changed:

Item	Finding	Change
Optics, dose arithmetic, S1, S28	Upheld: an independent trace reproduces the note within 2 %	None
PR-002, S2, S3	Weakened: 47.3 mJ/cm² with the 0.93 factor is below the note's own 48 for "Met"; a stack of LED derating gives 37.5	0.93 factor into `sizing.py` (B10), derating stack (B11); PR-002 now at risk; new S33 (plausible); dose margin to Amish (O-3)
S21, laminar as the pessimistic limit	Refuted: the profile never develops, the inlet jet and the warm cap mix the water; 5 to 10 tanks in series give 31 to 39 mJ/cm²	S21 restated to the Reynolds number only; new S34 (plausible, tracer test at TRL 4); inlet distributor to Amish (O-4; decided 2026-10-25: side inlet kept, tracer test at TRL 4)
PR-003, S8	Upheld and sharpened: 15.8 mJ/cm² at 70 %/cm with the factor; S8 is 42.1 with the factor, not 45.2	S8 recomputed with the factor; the note no longer says the test-water dose is above Class B
PR-004, alarm threshold	Weakened: threshold from the unweighted RED; one failed LED would not trip it	Fixed under the standing instruction (D-S4); new S38
S17, board temperature	Weakened: a fully developed film coefficient (about 44) would give 53 to 54 °C	Answered with S35: over the first 18 mm of stainless bore the flow is developing (Sieder-Tate, about 490 W/(m² K)), so the 150 to 600 range holds; still-water case already covered by the cut-back (F6)
S12, S13, ring torque	Weakened: the 0.3 N·m dry window was too narrow; 316L would fail	Fix revised (D-S2: anti-seize, 0.21 N·m, 316 sheet)
S22, pressure drop	Weakened: a direct-acting valve near Kv 0.10 gives 0.59 bar	G9 added; S22 plausible; PR-008 at risk until the valve is chosen
Start-up with a normally closed valve	Missing claim, and a real failure: the unit could not start	Fixed under the standing instruction (D-S3); new S37
Water hammer from the unit's own valve	Missing claim	New S36: 2.5 bar in the PE inlet tube, 6.5 bar peak on a 4 bar limiter, within 8 bar
S7, idle pulse dose	Weakened: a 10 s pulse gives about 38 mJ/cm², not 19	Corrected
S9	Weakened: the flux was not printed by the calculation; the reverse path was left out	The trace is in `stem_check.py` (S9 first-run print); the fix covers either path
S25, S29	Weakened: the long-wavelength LED tail and the sensor-patch ray count	New S39 (plausible); the sub-nanoampere figures are already flagged as noisy in the note
S31, S32	Weakened: only the reactor was reviewed	Both rows extended to the flow sensor, valve, pressure sensor and fittings; DEC To confirm item 11
	Upheld (the reviewer notes the 9 % window margin, the cycle count over	Kept; the window margin and fatigue go to the design FMEA and the TRL 4 pressure test

Item	Finding	Change
S11, S14 to S16, S18 to S20, S23, S24, S26, S27	hours for LED life, and the Vf bin taking power to 24.7 W)	

The reviewer also counted a stale line in the verdict (four plausible rows of option A where there were three) and a requirement count one too low in the PRS summary; both are corrected in this version.

Sources

- Beck, S. E., H. Ryu, L. A. Boczek, J. L. Cashdollar, K. M. Jeanis, J. S. Rosenblum, O. R. Lawal and K. G. Linden. "Evaluating UV-C LED Disinfection Performance and Investigating Potential Dual-Wavelength Synergy." *Water Research* 109 (2017): 207 to 216. https://pmc.ncbi.nlm.nih.gov/articles/PMC6145099, accessed 2026-10-25 (S3, S6).
- US Code of Federal Regulations, 40 CFR 141.720(d)(1), UV dose table for Cryptosporidium, Giardia lamblia and virus inactivation credit. https://www.law.cornell.edu/cfr/text/40/141.720, accessed 2026-10-25 (S4, S5).
- US Code of Federal Regulations, 21 CFR 177.1550 (perfluorocarbon resins) and 177.2470 (polyoxymethylene copolymer). https://www.law.cornell.edu/cfr/text/21/177.1550 and https://www.law.cornell.edu/cfr/text/21/177.2470, accessed 2026-10-25 (S31).
- Momentive Technologies, fused quartz mechanical and thermal properties. https://www.momentivetech.com/materials/fused-quartz-materials/quartz-properties/mechanical-properties, cited in LMF-CAL-001 (S11, S20, C3).
- ISO 3506-1, mechanical properties of corrosion-resistant stainless steel fasteners (A4-70 proof stress 450 MPa) (S14).
- ASTM A240, stainless steel plate, sheet and strip (316 minimum 0.2 % proof stress 205 MPa) (S12, S15).
- US EPA, *Membrane Filtration Guidance Manual*, EPA 815-R-06-009, 2005 (B2); ASTM D6908, integrity testing of water filtration membrane systems (B4).
- Young, W. C., and R. G. Budynas. *Roark's Formulas for Stress and Strain*. 7th ed. New York: McGraw-Hill, 2002 (plate and ring bending, S11, S12, S16).
- LMF-CAL-001 v0.7, `docs/04-calcs/01-sizing.md` and `sizing.py`, for every row closed by the calculation note.